Voltage Drop Chart

How far each copper wire size can run before it loses 3 percent of the voltage, at common currents and voltages.

For a 3 percent drop, the longest one-way run is 0.03 times the voltage times the wire area, divided by the current and a constant. At 120 V, 12 AWG copper carrying 20 A reaches the 3 percent mark at 45 ft. The same wire on a 240 V circuit goes twice as far, because the allowed drop in volts is twice as large.

Maximum one-way length for 3% drop: 120 V single-phase

Wire size	5 A	10 A	15 A	20 A	30 A	40 A	50 A
14 AWG	114	57	38	—	—	—	—
12 AWG	182	91	60	45	—	—	—
10 AWG	289	144	96	72	48	—	—
8 AWG	460	230	153	115	76	57	46
6 AWG	732	366	244	183	122	91	73
4 AWG	1,164	582	388	291	194	145	116
3 AWG	1,468	734	489	367	244	183	146
2 AWG	1,851	925	617	462	308	231	185
1 AWG	2,335	1,167	778	583	389	291	233
1/0 AWG	2,946	1,473	982	736	491	368	294
2/0 AWG	3,714	1,857	1,238	928	619	464	371
3/0 AWG	4,682	2,341	1,560	1,170	780	585	468
4/0 AWG	5,905	2,952	1,968	1,476	984	738	590

Copper conductors, in feet, rounded down. A dash means that current is more than the conductor may carry, so no length is given.

Maximum one-way length for 3% drop: 240 V single-phase

Wire size	15 A	20 A	30 A	40 A	50 A	60 A	80 A	100 A
14 AWG	76	—	—	—	—	—	—	—
12 AWG	121	91	—	—	—	—	—	—
10 AWG	193	144	96	—	—	—	—	—
8 AWG	307	230	153	115	92	—	—	—
6 AWG	488	366	244	183	146	122	—	—
4 AWG	776	582	388	291	232	194	145	—
3 AWG	978	734	489	367	293	244	183	146
2 AWG	1,234	925	617	462	370	308	231	185
1 AWG	1,557	1,167	778	583	467	389	291	233
1/0 AWG	1,964	1,473	982	736	589	491	368	294
2/0 AWG	2,476	1,857	1,238	928	742	619	464	371
3/0 AWG	3,121	2,341	1,560	1,170	936	780	585	468
4/0 AWG	3,936	2,952	1,968	1,476	1,181	984	738	590

Copper conductors, in feet, rounded down. A dash means that current is more than the conductor may carry, so no length is given.

Maximum one-way length for 3% drop: 480 V three-phase

Wire size	20 A	30 A	50 A	60 A	80 A	100 A	150 A	200 A
14 AWG	—	—	—	—	—	—	—	—
12 AWG	210	—	—	—	—	—	—	—
10 AWG	334	222	—	—	—	—	—	—
8 AWG	532	354	212	—	—	—	—	—
6 AWG	845	563	338	281	—	—	—	—
4 AWG	1,345	896	538	448	336	—	—	—
3 AWG	1,695	1,130	678	565	423	339	—	—
2 AWG	2,138	1,425	855	712	534	427	—	—
1 AWG	2,696	1,797	1,078	898	674	539	—	—
1/0 AWG	3,402	2,268	1,361	1,134	850	680	453	—
2/0 AWG	4,289	2,859	1,715	1,429	1,072	857	571	—
3/0 AWG	5,407	3,604	2,162	1,802	1,351	1,081	720	540
4/0 AWG	6,818	4,545	2,727	2,272	1,704	1,363	909	681

Copper conductors, in feet, rounded down. A dash means that current is more than the conductor may carry, so no length is given.

How the values are calculated

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L = (0.03 × V × cmil) / (2 × 12.9 × I)  [single-phase]
L = (0.03 × V × cmil) / (1.732 × 12.9 × I)  [three-phase]
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Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

The K-factor method uses resistance only. It ignores reactance, which matters on large conductors and long runs, so the lengths for the biggest sizes are optimistic. It assumes copper at about 75°C and a load at the far end. The 3 percent figure is a recommendation in the NEC informational notes for a branch circuit or a feeder, not a requirement, though some equipment and some local codes do require it.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/voltage-drop-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.